IABAC CERTIFIED DATA SCIENTIST PROJECT
Employee Performance Prediction

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Project Summary

INX Future Inc. experienced declining employee performance indexes, leading to operational delivery escalations and an eight percentage point drop in client satisfaction. Management lacked data backed visibility into the root causes of underperformance, risking arbitrary actions that could harm staff morale and tarnish the organization five year standing as a Top 20 employer.

This study analyzed 1,200 employee records across 28 organizational variables. A systematic analytical workflow was executed, encompassing rigorous exploratory data analysis, parametric and non parametric outlier detection, multicollinearity screening via Variance Inflation Factor analysis, ordinal logistic regression modeling, and advanced supervised machine learning algorithms.

The final optimized soft voting ensemble model, blending fine tuned Random Forest and XGBoost architectures, achieved an overall predictive accuracy of 94.00 percent, a Weighted F1 score of 0.940, and a Macro F1 score of 0.890. Crucially, the model solved the minority class identification challenge, capturing 93.10 percent of underperforming staff (Performance Rating 2) and identifying top performers (Performance Rating 4) with 100 percent precision. The primary drivers influencing employee performance were identified as Environment Satisfaction, Last Salary Hike Percentage, Years Since Last Promotion, and Experience Years In Current Role.

1. Business Problem

Business Background

INX Future Inc. is a leading global technology solutions enterprise that has maintained a distinguished five year track record as a Top 20 best employer. In recent operational cycles, overall employee performance ratings began to degrade. This decline manifested in severe service delivery escalations across client facing units and an eight percentage point decline in external client satisfaction scores.

Problem Statement

Taking broad or punitive disciplinary measures against underperforming employees risks alienating top tier talent, damaging workforce morale, and undermining

organizational culture. Executive leadership lacked an empirical, data driven framework to diagnose why performance was slipping and how to proactively forecast employee performance levels.

Core Stakeholders and Decision Matrix

- Chief Executive Officer (Mr. Brain) and Executive Leadership: Project sponsors requiring high level strategic visibility into organizational health, predictive levers, and ROI of workforce investments.

- Department Heads and Delivery Managers: Operational stakeholders directly impacted by client escalations who require team level workload and retention insights.

- Human Resources Department: Administrative unit responsible for redesigning compensation bands, career progression pathways, and targeted employee support initiatives.

Key Business Questions

- What is the baseline distribution of employee performance ratings across INX Future Inc., and what proportion represents underperformance?

- How does employee performance systematically vary across departments and functional job roles?

- Which employee characteristics and organizational factors exhibit statistically significant associations with performance?

- How accurately can employee performance ratings be predicted using modern classification algorithms?

- What evidence based interventions should INX Future Inc. deploy, and how can their operational efficacy be evaluated over time?

2. Data Understanding and Quality Assessment

Dataset

The dataset comprises 1,200 unique employee records and 28 attributes. Data completeness validation confirmed that all columns contain zero missing or null entries across both numerical and categorical features.

- Employee Demographics: Age spans from 18 to 60 years (mean of 36.92 years, median of 36.00 years), reflecting a balanced mid career workforce. Distance From Home averages 9.17 units (median of 7.00 units, maximum of 29.00 units), displaying positive skewness.

- Organizational Tenure: Total industry work experience averages 11.33 years. Experience at INX Future Inc. averages 7.08 years with a maximum of 40.00 years. Substantial differences between the 75th percentile and maximum values in company tenure (10.00 versus 40.00 years), years in current role (7.00 versus 18.00 years), and years since last promotion (3.00 versus 15.00 years) confirm positive skewness and long tenured personnel.

- Compensation and Learning: Employee Hourly Rate spans symmetrically from 30 to 100 (mean 65.98, median 66.00). Last Salary Hike Percentage ranges from 11.00 percent to 25.00 percent (mean 15.22 percent). Employees attend an average of 2.79 training sessions annually.

- Sentiment and Engagement: Workplace satisfaction metrics (Environment Satisfaction, Job Involvement, Job Satisfaction, Relationship Satisfaction, and Work Life Balance) are bounded on a 1 to 4 ordinal scale, all displaying consistent medians of 3.00 and means between 2.72 and 2.74.

- Target Variable (Performance Rating): Scores are bounded between 2.00 and 4.00. Rating 1 is entirely absent from the dataset. Ratings exhibit severe class imbalance:

    - Rating 3 (Standard Performers): 874 employees (72.83 percent)
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    - Rating 2 (Underperforming Staff): 194 employees (16.17 percent)
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    - Rating 4 (Top Tier Performers): 132 employees (11.00 percent)

Outlier Detection

Both non parametric (Interquartile Range) and parametric (Z score cutoff of 3.0) methodologies were applied to numerical features. Tenured career variables (such as Total Work Experience In Years, Experience Years At This Company, and Years Since Last Promotion) exhibited right tail distributions. Because these values represent genuine, legitimate senior employee career paths rather than measurement entry errors, records were retained to avoid removing institutional history.

3. Exploratory Data Analysis and Cross Tabulation

Performance Across Departments

Descriptive cross tabulation revealed marked differences in performance distributions across corporate departments:

- Finance: Displayed the highest concentration of underperformance, with 30.61 percent of staff receiving Performance Rating 2.

- Sales: Recorded 23.32 percent of its personnel in the Performance Rating 2 bracket.

- Development: Achieved the highest proportion of top performers, with 12.19 percent rated 4 and only 3.60 percent rated 2.
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- Data Science: Maintained an average performance rating of 3.05, representing one of the highest technical baselines across the organization.

Performance Across Job Roles

Functional roles showed wide variation in performance distributions:

- High Underperformance Concentration: Finance Managers recorded the highest proportion of Rating 2 evaluations (30.61 percent), followed by Healthcare Representatives (24.24 percent) and Sales Executives (23.70 percent).

- High Performance Concentration: Managers achieved the highest proportion of Rating 4 evaluations (19.61 percent), followed by Business Analysts (18.75 percent) and Research Scientists (18.18 percent).

- Standard Concentration: Delivery Managers and Technical Architects were clustered exclusively at Performance Rating 3 (100 percent), with zero staff rated either 2 or 4.

4. Econometric and Statistical Significance Modeling

Multicollinearity Diagnostic (Variance Inflation Factor)

Prior to statistical inference, multicollinearity was diagnosed across all independent variables using the Variance Inflation Factor (VIF). Features displaying perfect or near perfect linear combinations (Attrition, Overtime, and redundant Department indicators) were pruned from the explanatory framework.

All continuous and ordinal predictors demonstrated stable VIF figures below the conservative threshold of 5.0:

- Total Work Experience In Years: VIF = 4.98
- Experience Years At This Company: VIF = 4.75
- Employee Job Level: VIF = 3.30
- Experience Years In Current Role: VIF = 2.92

Elevated VIF metrics observed in select dummy categories (such as Education Background Life Sciences at 21.04 and Medical at 19.07) were verified as structural mathematical artifacts of small reference classes rather than problematic collinearity.

Pruned Ordinal Logistic Regression Results
Because the target labels maintain a distinct progressive order (Rating 2 < Rating 3 < Rating 4), an Ordered Logit model was fitted on 960 training observations across nine pruned predictors:

- Employee Environment Satisfaction: Coefficient = 1.2328, z = 13.360, $p < 0.001$ (Strongest positive determinant)
- Employee Last Salary Hike Percent: Coefficient = 1.1395, z = 12.084, $p < 0.001$ (Highly significant positive determinant)
- Employee Work Life Balance: Coefficient = 0.4588, z = 3.954, $p < 0.001$ (Positive determinant)
- Experience Years At This Company: Coefficient = 0.3705, z = 2.801, $p = 0.005$ (Positive determinant)
- Experience Years In Current Role: Coefficient = negative 0.5855, z = negative 4.636, $p < 0.001$ (Significant negative relationship)
- Years Since Last Promotion: Coefficient = negative 0.3986, z = negative 3.948, $p < 0.001$ (Significant negative relationship)
- Employee Job Role Finance Manager: Coefficient = negative 1.1791, z = negative 2.921, $p = 0.003$ (Negative role baseline effect)

- Employee Education Level: Coefficient = 0.1376, z = 1.773, p = 0.076 (Marginally positive effect)

- Education Background Medical: Coefficient = negative 0.1420, z = negative 0.821, p = 0.412 (Not statistically significant)

5. Machine Learning Benchmarking and Imbalance Remediation
Data Preprocessing Pipeline

- Data Splitting: An 80:20 stratified split partitioned the 1,200 records into 960 training instances and 240 unseen test instances, strictly preserving target class ratios.

- Encoding: Ordinal ratings were mapped onto integer scales. Nominal features (Education Background, Job Role, Marital Status) were dummy encoded using k minus 1 binaries.

- Feature Scaling: Continuous variables were normalized using StandardScaler to zero mean and unit variance.

Class Imbalance Remediation Strategies
To address the heavy concentration of Rating 3 (72.83 percent), four distinct training formulations were tested using standard Logistic Regression:

Method	Accuracy	Macro F1	Weighted F1	Rating 2 Recall	Rating 4 Recall	
1. Baseline Unweighted	0.842	0.736	0.842	0.552	0.704	
2. Class Weight Balanced	0.746	0.666	0.76	0.828	0.741	
3. Standard SMOTE	0.767	0.685	0.781	0.655	0.704	

Diagnostic Takeaway: While SMOTENC improved minority recall over standard SMOTE by respecting categorical boundaries, linear models suffered overall accuracy degradation when trying to compensate for minority categories.

Supervised Model Comparison Matrix

Non linear tree ensembles were subsequently trained and benchmarked against the linear baseline across identical 240 holdout test samples:

Model Architecture	Accuracy	Macro F1	Weighted F1	Rating 2 Recall	Rating 4 Recall
1. Logistic Regression Base	0.842	0.736	0.842	0.552	0.704
2. Balanced Logistic Regression	0.746	0.666	0.76	0.828	0.741
3. Unweighted Random Forest	0.925	0.87	0.923	0.931	0.667
4. Balanced Random Forest	0.912	0.858	0.915	0.828	0.704
5. Unweighted XGBoost	0.908	0.847	0.907	0.828	0.741
6. Balanced XGBoost	0.921	0.871	0.921	0.897	0.778
7. Baseline Soft Voting (RF+XGB)	0.929	0.881	0.929	0.897	0.815

8. Final Tuned Ensemble	0.94	0.89	0.94	0.931	0.741

Diagnostic Takeaway: The leap in performance from linear classifiers (84.20 percent) to tree ensembles (above 92.00 percent) confirms that employee performance is governed by non linear decision thresholds and multi feature interactions that linear equations cannot capture.

6. Hyperparameter Tuning and Final Architecture

Hyperparameter Optimization

Both candidate algorithms were subjected to RandomizedSearchCV utilizing three fold stratified cross validation scored on Macro F1:

1. Tuned Random Forest: n estimators = 200, max depth = 20, min samples split = 2, min samples leaf = 1.
2. Tuned XGBoost: n estimators = 200, learning rate = 0.05, max depth = 4, subsample = 0.80, eval metric = mlogloss.

Probability Blending and Ensemble Formulation

Predicted probability distributions from the best estimators were blended via a soft voting grid search. The optimal weight distribution was established at 0.90 for Random Forest and 0.10 for XGBoost, demonstrating that deeply tuned bagging provided the core classification stability while boosting smoothed probabilities near complex decision boundaries.

Final Holdout Diagnostic (Tuned Soft Voting Ensemble)

Evaluated across 240 test instances (Rating 2: 29 cases, Rating 3: 184 cases, Rating 4: 27 cases):

Performance Class	Precision	Recall	F1 Score	Support
Rating 2 (Low)	0.82	0.93	0.87	29
Rating 3 (Standard)	0.96	0.96	0.96	184

Rating 4 (High)	1	0.74	0.85	27
Metric	**Value**			
Overall Accuracy	94.00%			
Macro Average F1	0.89			
Weighted Average F1	0.94			

Diagnostic Findings: The tuned ensemble correctly classified 226 out of 240 test records. Crucially, the model produced zero extreme classification errors (zero Rating 2 employees misclassified as Rating 4, and zero Rating 4 employees misclassified as Rating 2).

7. Feature Importance and Root Cause Insights

Relative Gini impurity metrics extracted from the tuned ensemble identified the top performance drivers across the workforce:

- Environment Satisfaction (Importance: 0.14): Represents the single largest determinant of whether an employee meets expectations or slips into underperformance.
- Last Salary Hike Percentage (Importance: 0.12): Reflects compensation competitiveness and recognition for output. Stagnant compensation closely precedes drops in productivity.
- Years Since Last Promotion (Importance: 0.09): Extended periods without upward mobility trigger career fatigue and disengagement.
- Experience Years In Current Role (Importance: 0.08): Employees left in identical job roles without expanded scope or rotation experience lower motivation.

- Experience Years At This Company (Importance: 0.07): Institutional familiarity supports baseline competence, but tenure without career progression degrades output.

- Department Context (Importance: 0.06): Departmental culture, leadership quality, and workload distributions create baseline performance variations across functional units.

8. Strategic Recommendations for INX Future Inc.

Priority 1: Workplace Environment and Engagement Modernization

- Insight: Environment satisfaction is the leading predictor of employee performance across all departments.

- Action: Institutionalize monthly departmental pulse surveys and feedback mechanisms to uncover operational bottlenecks, tooling deficiencies, and managerial friction points. Empower department heads to address local culture issues before sentiment declines.

Priority 2: Transparent Merit Based Compensation Frameworks

- Insight: Salary hike percentage strongly separates high performers from underperforming staff.

- Action: Conduct an immediate compensation audit against current industry market bands. Realign compensation policies so that salary adjustments transparently reflect objective KPI milestones, ensuring strong contributors are rewarded before motivation drops.
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Priority 3: Career Progression Pathways and Rotation Architecture

- Insight: High role tenure coupled with stagnant promotion intervals creates severe performance deterioration.

- Action: Establish structured career mapping frameworks and internal mobility programs. For employees who have spent more than three years in a role without a formal title change, provide lateral cross departmental rotations, leadership training, or role redesigns.

Priority 4: Functional Knowledge Transfer from High Performing Units

- Insight: Development (average rating 3.09) and Data Science (average rating 3.05) maintain superior performance distributions compared to Finance (average rating 2.78).

- Action: Study the agile collaboration frameworks, technical workflows, and team culture within Development and Data Science. Transfer applicable delivery and mentorship practices into Finance and Sales to modernize team workflows.

9. Quantitative Evaluation of Interventions

To measure whether corporate interventions achieve the executive mandate of lifting performance and improving client satisfaction by eight percentage points, INX Future Inc. should execute three empirical tracking mechanisms:

Mechanism 1: Longitudinal Rating Migration Analysis

- Execution: Process quarterly employee feature datasets through the serialized prediction pipeline.

- Success Metric: Successful policy rollouts must yield a quarter over quarter percentage decline in the volume of personnel flagged as Performance Rating 2, paired with a proportional upward migration into Performance Ratings 3 and 4.

Mechanism 2: Cohort Based Controlled A/B Pilots

- Execution: Deploy specific interventions (such as updated compensation schedules or flexible scheduling) to select pilot departments while maintaining other units as control groups.

- Success Metric: Compare predicted performance probability distributions between pilot and control groups after two business quarters. A statistically significant rightward shift in predicted performance probabilities within the pilot group confirms intervention efficacy.

Mechanism 3: Feature Importance Sensitivity Re-Estimation

- Execution: Retrain and extract feature importance metrics semi annually.

- Success Metric: Effective organizational interventions targeting specific friction points (such as aggressive compensation restructuring) will cause the model predictive reliance on Last Salary Hike Percentage to drop, indicating that compensation is no longer an active performance bottleneck.